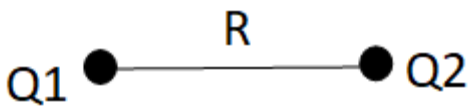
MID – 1 Question Bank Answers

UNIT - I

1. (a) State and explain coulomb's law? Obtain an expression in vector form?

Ans:

Coulomb's law states that the force F' between two point charges Q_1 and *Q2* is:
1. Along the line joining them
2. Directly proportional to the product Q_1Q_2 of the charges
3. Inversely proportional to the square of the distance *R* between them



Expressed mathematically as

$$\mathbf{F} \propto \frac{\mathbf{Q_1 Q_2}}{\mathbf{R^2}}$$

$$\mathbf{F} = \mathbf{K}\frac{\mathbf{Q_1 Q_2}}{\mathbf{R^2}}$$

Where *k* is the proportionality constant. In SI units, charges Q1 and Q2 are in coulombs (C), The distance R is in meters (m), and the force F is in newtons (N) so that $\boldsymbol{K} = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_o}}$

The constant $\boldsymbol{\varepsilon_o}$ is known as the permittivity of free space and $\boldsymbol{\varepsilon_o}$ = 8.854 X 10^{-12} F/m.

Therefore K=9X 10^9 m/F.

Vector form of Coulomb's law :

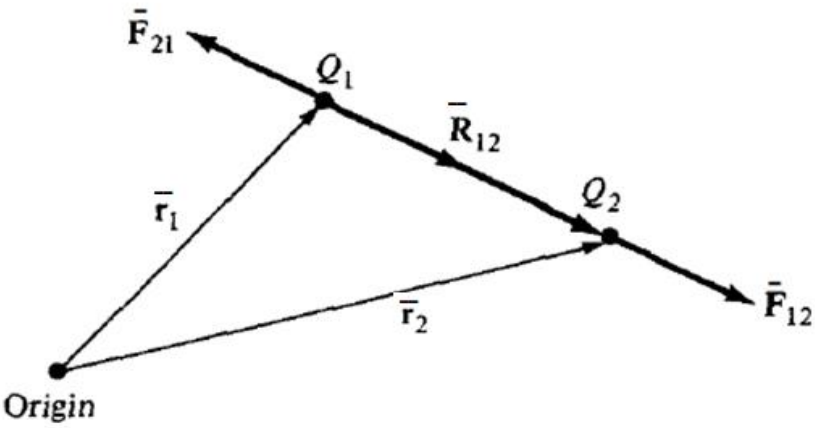


Fig. Vector form of Coulomb's law

If point charges Q_1 and Q_2 are located at points having position vectors r_1 and r_2, then the force $\bar{F}_{12}$ on Q_2 due to Q_1 is given by

$$\bar{\mathbf{F}}_{\mathbf{12}} = \frac{\mathbf{Q_1 Q_2}}{\mathbf{4\pi\varepsilon_o R^2}}\bar{\mathbf{a}}_{\mathbf{R_{12}}}$$

Where $\bar{R}_{12} = \bar{r}_2 - \bar{r}_1$

$$R = |\bar{R}_{12}| = |\bar{r}_2 - \bar{r}_1|$$

$$\bar{a}_{R_{12}} = \frac{\bar{R}_{12}}{R}$$

$$\bar{F}_{12} = \frac{Q_1 Q_2}{4\pi\varepsilon_o R^3}\bar{R}_{12}$$

$$\bar{F}_{12} = \frac{Q_1 Q_2(\bar{r}_2 - \bar{r}_1)}{4\pi\varepsilon_o|\bar{r}_2 - \bar{r}_1|^3}$$

Similarly the force F_{21}, on Q_1 due to Q2 is given by

$$\bar{F}_{21} = -\bar{F}_{12}$$

1. b) A point charge, Q1 = 2μC is at (2, 3, 6) and another charge, Q2 = 5μC is at (0, 0, 0) in free space. Find the force on Q1 due to Q2.

Ans:

Given that $Q_1 = 2\mu C$ and $Q_2 = 5\mu C$

$P_1(2,3,6)$ and $P_2(0,0,0)$

$\bar{r}_1 = 2\bar{a}_x + 3\bar{a}_y + 6\bar{a}_z$ and $\bar{r}_2 = 0\bar{a}_x + 0\bar{a}_y + 0\bar{a}_z$

$\bar{R}_{21} = \bar{r}_1 - \bar{r}_2 = 2\bar{a}_x + 3\bar{a}_y + 6\bar{a}_z$

$R_{21} = |\bar{R}_{21}| = \sqrt{2^2 + 3^2 + 6^2} = \sqrt{4+9+36} = \sqrt{49} = 7$

$\bar{a}_{R_{21}} = \frac{\bar{R}_{21}}{|\bar{R}_{21}|} = \frac{2\bar{a}_x + 3\bar{a}_y + 6\bar{a}_z}{7}$

Force on Q_1 due to Q_2 is given by

$$\bar{F}_{21} = \frac{Q_1 Q_2}{4\pi\varepsilon_o R_{21}^2} \cdot \bar{a}_{R_{21}}$$

$$= \frac{2\times10^{-6}\times5\times10^{-6}}{4\pi\times8.854\times10^{-12}\times7^2} \cdot \frac{2\bar{a}_x + 3\bar{a}_y + 6\bar{a}_z}{7}$$

$$= \frac{10\times10^{-12}(2\bar{a}_x + 3\bar{a}_y + 6\bar{a}_z)}{4\pi\times8.854\times7^3\times10^{-12}}$$

$$= \frac{10(2\bar{a}_x + 3\bar{a}_y + 6\bar{a}_z)}{38163}$$

$$= 0.262\times10^{-3}(2\bar{a}_x + 3\bar{a}_y + 6\bar{a}_z)$$

$$= (0.524\bar{a}_x + 0.786\bar{a}_y + 1.572\bar{a}_z)\times10^{-3}\ N$$

$$\boxed{\therefore \bar{F}_{21} = 0.524\bar{a}_x + 0.786\bar{a}_y + 1.572\bar{a}_z\ mN}$$

2. (a) State and explain Gauss Law.

Ans:

Gauss's law states that the total electric flux ψ through any closed surface is equal to the total charge enclosed by that surface.

$$\boxed{\psi = Q} \quad ---(1)$$

$$\text{Total flux } \psi = \oint_s \bar{D}\cdot\bar{ds} \quad ---(2)$$

$$\text{Total charge enclosed } Q = \int_v \rho_v dv \quad ---(3)$$

Substitute the Eq.(2) and Eq.(3) in equation (1), we get

$$\boxed{\oint_s \bar{D} \cdot \overline{ds} = \int_v \rho_v dv} \qquad ----(4)$$

This equation is called integral form of Gauss's law

By applying divergence theorem to the first term in the above equation

$$\oint_s \bar{D} \cdot \overline{ds} = \int_v \nabla \cdot \bar{D} dv$$

$$\int_v \nabla \cdot \bar{D} dv = \int_v \rho_v dv$$

$$\boxed{\nabla \cdot \bar{D} = \rho_v}$$

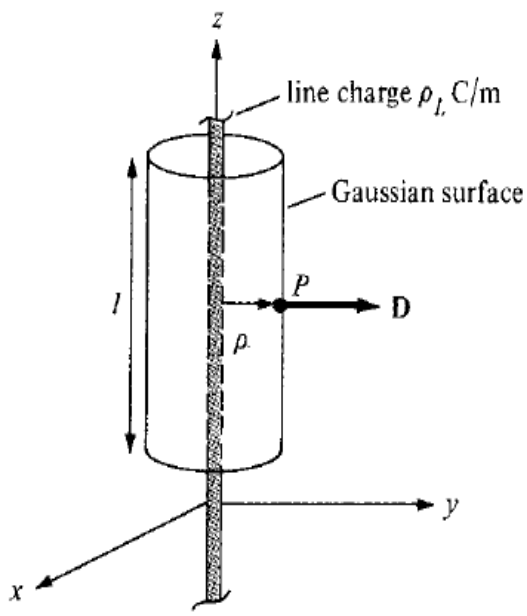
The above equation is called point form of gauss's law.

2. (b) Determine an expression for the electric field E due to infinite line charge using Gauss Law.

Ans:

<u>Electric field intensity due to Infinite Line Charge:</u>

Let us consider the infinite line of uniform charge ρ_L C/m lies along the z-axis.



- Fig. Fig. Gaussian surface about an infinite line charge

To determine D at a point P, construct the cylindrical surface with radius ρ around infinite line charge as Gaussian surface.

From Gauss's law we know that $\oint_s \bar{D} \cdot \overline{ds} = Q$

$$\oint_s D\, dS = Q$$

$$D \oint_s dS = Q$$

Where $\oint_s dS = 2\pi\rho l$ is the surface area of the Gaussian surface.

$\oint_s \bar{D} \cdot \overline{ds}$ evaluated on the top and bottom surfaces of the cylinder is zero, since D is tangential to the top and bottom surfaces.

$$D(2\pi\rho l) = \rho_L\, l$$

$$\mathbf{D} = \frac{\rho_L}{2\pi\rho}$$

The electric flux density is

$$\bar{\mathbf{D}} = \frac{\rho_L}{2\pi\rho}\bar{\mathbf{a}}_\rho$$

Electric field intensity due to infinite line charge is

$$\bar{\mathbf{E}} = \frac{\rho_L}{2\pi\varepsilon_0\rho}\bar{\mathbf{a}}_\rho$$

3. a) Determine an expression for the electric field E due to infinite charge sheet using Gauss Law.

Ans:

<u>Electric field intensity due to Infinite Sheet of Charge:</u>

Consider the infinite sheet of uniform charge ρ_s C/m^2 lying on the $z = 0$ plane as shown in fig.

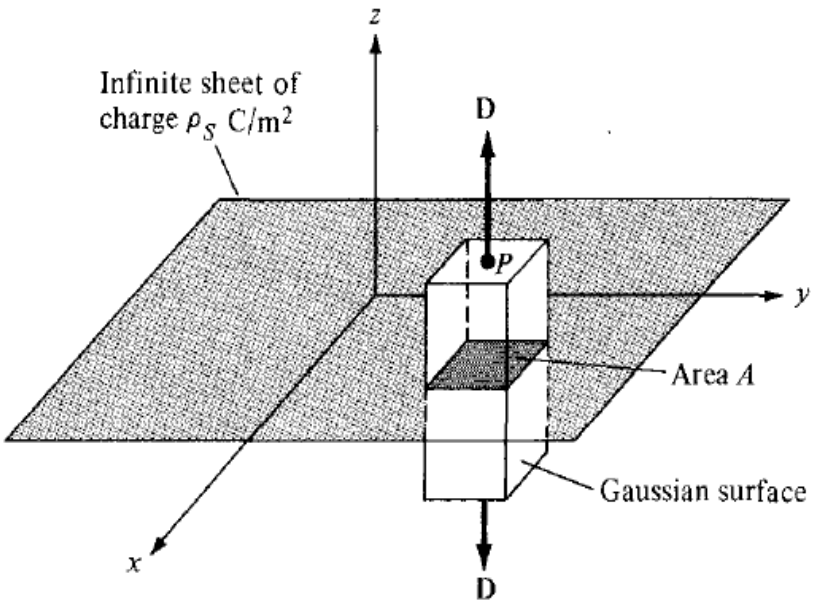


Fig. An infinite sheet of charge.

To determine E at a point P, construct rectangular box as shown in fig.

From Gauss's law we know that

$$\oint_s \bar{\mathbf{D}} \cdot \overline{\mathbf{ds}} = \mathbf{Q}$$

$$\oint_{top} \bar{\mathbf{D}} \cdot \overline{\mathbf{ds}} + \oint_{bottom} \bar{\mathbf{D}} \cdot \overline{\mathbf{ds}} = \mathbf{Q}$$

Note that $\bar{\mathbf{D}} \cdot \overline{\mathbf{ds}}$ evaluated on the sides of the box is zero because D is tangential to the sides.

$$\oint_{top} \mathbf{DdS} + \oint_{bottom} \mathbf{DdS} = \mathbf{Q}$$

$$\mathbf{D}\left(\oint_{top} \mathbf{dS} + \oint_{bottom} \mathbf{dS}\right) = \mathbf{Q}$$

If the top and bottom area of the box each has area A, then above equation becomes

$$\mathbf{D(A + A) = \rho_S A}$$

$$\mathbf{D2A = \rho_S A}$$

$$D = \frac{\rho_S}{2}$$

$$\bar{D} = \frac{\rho_S}{2}\bar{a}_Z$$

Electric field intensity due to infinite sheet charge is

$$\bar{E} = \frac{\rho_S}{2\varepsilon_0}\bar{a}_Z$$

3. (b) If an electric field in free space is given by E = $a_x + 2a_y + 5a_z$ V/m, find the Electric flux density and $\epsilon_r = 4$.

Ans:

Relationship between $\bar{E}$ and $\bar{D}$ is given by

$$\bar{D} = \varepsilon \bar{E}$$

where $\varepsilon = \varepsilon_r \varepsilon_0 = 4 \times \varepsilon_0 = 4 \times 8.854 \times 10^{-12} = 35.416 \times 10^{-12}$

Now, $\bar{D} = \varepsilon \bar{E}$

$$= 35.416 \times 10^{-12} (\bar{a}_x + 2\bar{a}_y + 5\bar{a}_z)$$

$$= 0.035416 \times 10^{-9} (\bar{a}_x + 2\bar{a}_y + 5\bar{a}_z)$$

$$= (0.035416\,\bar{a}_x + 0.070832\,\bar{a}_y + 0.17708\,\bar{a}_z) \times 10^{-9}\ C/m^2$$

$$\therefore \bar{D} = (0.035416\,\bar{a}_x + 0.070832\,\bar{a}_y + 0.17708\,\bar{a}_z)\ nC/m^2$$

4. Derive two Maxwell's equations for electrostatic field.

Ans:

The two Maxwell's equations for the electrostatic field are:

1. Gauss's law:

Gauss's **law** states that the total electric flux ψ passing through any *closed* surface is equal to the total charge enclosed by that surface.

$$\boxed{\psi = Q} \quad \text{----------(1)}$$

$$\text{Total flux } \psi = \oint_s \bar{D} \cdot \overline{ds} \quad \text{----------(2)}$$

$$\text{Total charge enclosed } Q = \int_v \rho_v dv \quad \text{-----(3)}$$

$$\boxed{\oint_s \bar{D} \cdot \overline{ds} = \int_v \rho_v dv} \quad \text{----------(4)}$$

The equation is called integral form of Gauss's law.

By applying divergence theorem to the first term in eq. (4) i.e.,

$$\oint_s \bar{D} \cdot \overline{ds} = \int_v \nabla \cdot \bar{D} dv$$

Therefore

$$\int_v \nabla \cdot \bar{D} dv = \int_v \rho_v dv$$

$$\boxed{\nabla \cdot \bar{D} = \rho_v}$$

- The above equation is called point form of gauss's law.
- This equation is known as Maxwell's First Equation.

<u>2. Conservative nature of electrostatic field :</u>

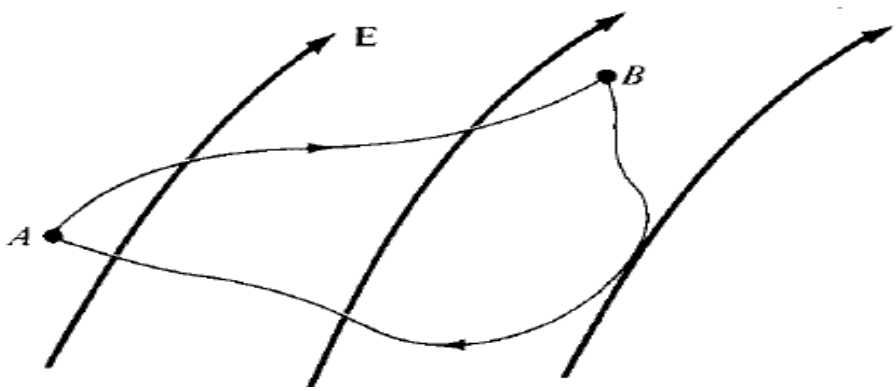


Figure: Conservative nature of an electrostatic field

- The potential difference between points *A and B is independent* of the path taken. Hence,

$$V_{AB} = -V_{BA}$$

- That is

$$V_{AB} + V_{BA} = \oint \bar{E} \cdot \overline{dl} = 0$$

Or

$$\oint \bar{E} \cdot \overline{dl} = 0 \quad ----(1)$$

- Eq(1) shows that the line integral of E along a closed path as shown in Fig must be zero.
- Applying Stokes's theorem to eq. (1) gives

$$\oint \bar{E} \cdot \overline{dl} = \int_s (\nabla \times \bar{E}) \cdot \overline{ds} = 0$$

Or

$$\nabla \times \bar{E} = 0 \quad \text{-----}(2)$$

- A field whose line integral does not depend on the path of integration is called conservative field hence, Electrostatic field is conservative field.
- Eq.(2) is also called Maxwell's Second equation.

5. a) Derive the Poisson and Laplace Equations and mention their Applications.

Ans:

Poisson's and Laplace's equations are easily derived from Gauss's law (for a linear medium)

$$\nabla \cdot \bar{D} = \rho_v$$

$$\nabla \cdot \varepsilon \bar{E} = \rho_v \quad \text{--------}(1)$$

$$\text{But} \quad \bar{E} = -\nabla V \quad \text{---------}(2)$$

Substitute eq. (2) in eq. (1), we get

$$\nabla \cdot \varepsilon(-\nabla V) = \rho_v$$

$$-\varepsilon \nabla^2 V = \rho_v$$

$$\boxed{\nabla^2 V = -\frac{\rho_v}{\varepsilon}}$$

This is known as Poisson's equation.

- When $\rho_v = 0$ (charge-free region), the above equation then becomes

$$\boxed{\nabla^2 V = 0}$$

which is known as Laplace's equation.

Applications:

Laplace Equation: No sources, used in empty regions or steady-state systems.

Poisson Equation: With sources, used to solve for fields or potentials due to distributions of charge, heat, or mass.

5. b) Derive the relationship between Electric Field and Electric Potential.

Ans:

From the definition of potential,

$$V = -\int \bar{E} \cdot \bar{dl}$$

It follows that

$$dV = -\bar{E} \cdot \bar{dl}$$

Where

$$\bar{E} = E_x \bar{a}_x + E_y \bar{a}_y + E_z \bar{a}_z$$

$$\bar{dl} = dx\, \bar{a}_x + dy\, \bar{a}_y + dz\, \bar{a}_z$$

$$dV = -E_x dx - E_y dy - E_z dz \text{ ----------(1)}$$

But from calculus of multiple variables

$$dV = \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz \text{ ----------(2)}$$

Comparing the two expressions for *dV*, we obtain

$$E_x = -\frac{\partial V}{\partial x}, \qquad E_y = -\frac{\partial V}{\partial y}, \qquad E_z = -\frac{\partial V}{\partial z}$$

Thus

$$\bar{E} = -\frac{\partial V}{\partial x}\bar{a}_x - \frac{\partial V}{\partial y}\bar{a}_y - \frac{\partial V}{\partial z}\bar{a}_z$$

$$\boxed{\bar{E} = -\nabla V}$$

That is, the electric field intensity is the -ve gradient of V.

6. Define capacitance. Derive the capacitance of a parallel plate capacitor.

Ans:

CAPACITANCE:

- The capacitance C of the capacitor is defined as the ratio of the magnitude of the charge on one of the plates to the potential difference between them. that is,

$$\boxed{C = \frac{Q}{V} = \frac{\oint_S \bar{D} \cdot d\bar{s}}{-\int \bar{E} \cdot d\bar{l}}}$$

The capacitance is measured in farads (F).

Parallel-Plate Capacitor:

- Consider the parallel-plate capacitor as shown in fig.
- Let S be area of each plate and d be the distance between two plates.
- We assume that plates 1 and 2 carry charges +Q and -Q respectively so that

$$\rho_s = \frac{Q}{S}$$

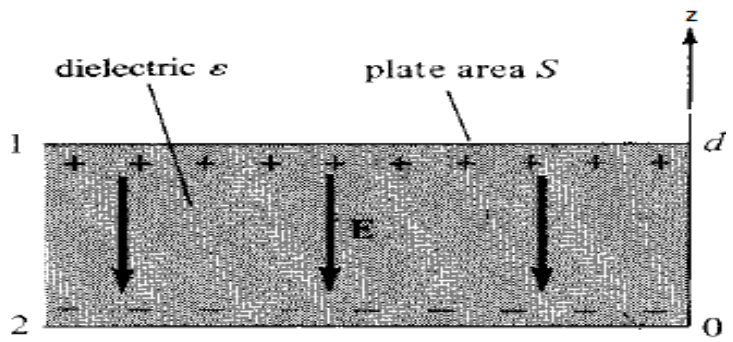


Figure: Parallel-plate capacitor

- If the space between the plates is filled with a homogeneous dielectric with permittivity ε, then electric field intensity inside the plates is given by

$$\bar{E} = -\frac{\rho_s}{\varepsilon}\,\bar{a}_z$$

$$\bar{E} = -\frac{Q}{\varepsilon S}\,\bar{a}_z$$

- The potential difference between the plates is given by

$$V = -\int_2^1 \bar{E}\cdot d\bar{l}$$

$$V = -\int_0^d \left[-\frac{Q}{\varepsilon S}\,\bar{a}_z\right]\cdot dz\bar{a}_z$$

$$V = \int_0^d \frac{Q}{\varepsilon S}\,dz$$

$$V = \frac{Q}{\varepsilon S}\int_0^d dz$$

$$V = \frac{Qd}{\varepsilon S}$$

- The capacitance for a parallel-plate capacitor is given by

$$C = \frac{Q}{V}$$

$$\boxed{C = \frac{\varepsilon S}{d}}$$

- The energy stored in a capacitor is given by

$$\boxed{W_E = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C}}$$

UNIT – II

1. (a) State and explain Biot-Savart Law.

Ans:

BIOT-SAVART'S LAW:

Biot-Savart's law states that the magnetic field intensity dH produced at a point P, as shown in Fig by the differential current element Idl is proportional to the product Idl and the sine of the angle α between the element and the line joining P to the element and is inversely proportional to the square of the distance R between P and the element.

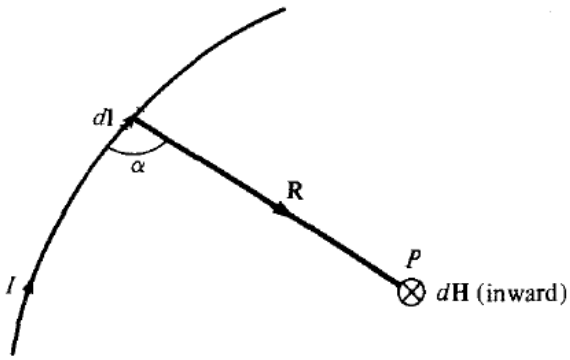


Fig. Magnetic field dH at P due to current element I dl.

That is $$\mathbf{dH} \propto \frac{\mathbf{Idl\ \sin\alpha}}{\mathbf{R^2}}$$

$$\mathbf{dH} = \mathbf{K}\frac{\mathbf{Idl\ \sin\alpha}}{\mathbf{R^2}}$$

Where k is the constant of proportionality.

In SI units, k = l/4π, so above equation becomes

$$\mathbf{dH} = \frac{\mathbf{Idl\ \sin\alpha}}{\mathbf{4\pi R^2}}$$

From the definition of cross product the above equation becomes

$$\boxed{\mathbf{d\bar{H}} = \frac{\mathbf{Id\bar{l}\times\bar{a}_R}}{\mathbf{4\pi R^2}} = \frac{\mathbf{Id\bar{l}\times\bar{R}}}{\mathbf{4\pi R^3}}}$$

Where $R = |\bar{R}|$ and $\bar{a}_R = \frac{\bar{R}}{R}$

1. (b) Derive the magnetic field intensity due to an Infinite line placed along z-axis at an observation point P on y axis.

Ans:

Let us consider an infinite line conductor that carries current I and lies along the z-axis as in Fig.

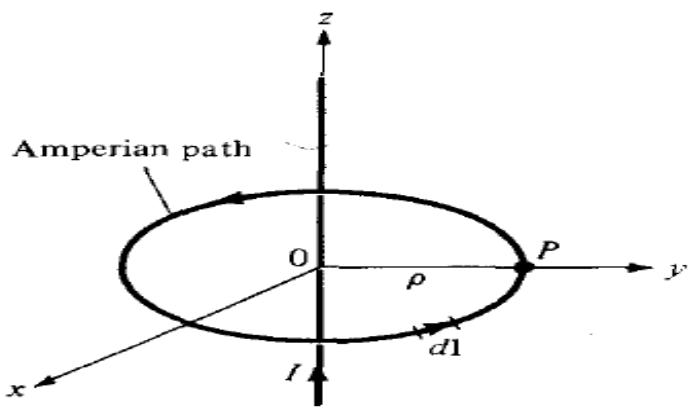


Fig. Ampere's law applied to an infinite line current.

To determine H at point P, construct a closed path passing through P around the conductor. It is also called as an Amperian path.
According to ampere's law $\oint \bar{\mathbf{H}}.\mathbf{d}\bar{\mathbf{l}} = \mathbf{I}$ ---------- (1)
H has only H_ϕ component and $\bar{\mathbf{d}}\mathbf{l} = \boldsymbol{\rho}\mathbf{d}\boldsymbol{\phi}\bar{\mathbf{a}}_{\boldsymbol{\phi}}$
Therefore $\int_0^{2\pi} \mathbf{H}_{\boldsymbol{\phi}}\bar{\mathbf{a}}_{\boldsymbol{\phi}} \cdot \boldsymbol{\rho}\mathbf{d}\boldsymbol{\phi}\bar{\mathbf{a}}_{\boldsymbol{\phi}} = \mathbf{I}$

$$\int_0^{2\pi} \mathbf{H}_{\boldsymbol{\phi}}\boldsymbol{\rho}\mathbf{d}\boldsymbol{\phi} = \mathbf{I}$$

$$\mathbf{H}_{\boldsymbol{\phi}}\boldsymbol{\rho}\int_0^{2\pi} \mathbf{d}\boldsymbol{\phi} = \mathbf{I}$$
$$\mathbf{H}_{\boldsymbol{\phi}}\boldsymbol{\rho}\mathbf{2}\boldsymbol{\pi} = \mathbf{I}$$
$$\mathbf{H}_{\boldsymbol{\phi}} = \frac{\mathbf{I}}{\mathbf{2}\boldsymbol{\pi}\boldsymbol{\rho}}$$

The magnetic field intensity is given by

$$\boxed{\bar{\mathbf{H}} = \frac{\mathbf{I}}{\mathbf{2}\boldsymbol{\pi}\boldsymbol{\rho}}\bar{\mathbf{a}}_{\boldsymbol{\phi}}}$$

2. (a) State and explain the ampere's circuital law.

Ans:

Ampere's circuit law states that the line integral of H around a closed path is the same as the net current I enclosed by the path.
In other words, the circulation of H equals I, that is,

$$\boxed{\oint \bar{\mathbf{H}}.\mathbf{d}\bar{\mathbf{l}} = \mathbf{I}} \qquad \text{---------- (1)}$$

By applying Stoke's theorem to the left-hand side of eq. (1), we obtain

$$\oint \bar{\mathbf{H}}.\mathbf{d}\bar{\mathbf{l}} = \int_S (\boldsymbol{\nabla} \times \bar{\mathbf{H}}).\mathbf{d}\bar{\mathbf{S}} \qquad \text{---------- (2)}$$

But

$$\mathbf{I} = \int_S \bar{\mathbf{J}}.\mathbf{d}\bar{\mathbf{S}} \qquad \text{---------- (3)}$$

Substitue the equations (2) and (3) in equation (1) then we get

$$\int_S (\boldsymbol{\nabla} \times \bar{\mathbf{H}}).\mathbf{d}\bar{\mathbf{S}} = \int_S \bar{\mathbf{J}}.\mathbf{d}\bar{\mathbf{S}}$$

Finally

$$\boxed{\boldsymbol{\nabla} \times \bar{\mathbf{H}} = \bar{\mathbf{J}}}$$

This is ampere's law in point form. This is the third Maxwell's equation.

2. (b) State and explain Faraday's law of induction. And hence introduce the concept of time varying electromagnetic fields through this law.

Ans:

<u>FARADAY'S LAW:</u> Faraday law states that the induced emf V_{emf} (in volts), in any closed circuit is equal to the time rate of change of the magnetic flux linkage by the circuit.

$$\mathbf{V}_{\mathbf{emf}} = -\mathbf{N}\frac{\mathbf{d}\boldsymbol{\psi}}{\mathbf{dt}}$$

Where N is the number of turns in the circuit and ψ is the flux through each turn. The negative sign shows that the induced voltage acts in such a way as to oppose the flux producing it. This is known as Lenz's law.
For a circuit with a single turn (N = 1), then above equation becomes

$$\mathbf{V}_{\mathbf{emf}} = -\frac{\mathbf{d}\boldsymbol{\psi}}{\mathbf{dt}} \qquad ------(\mathbf{1})$$

Induced e.m.f in the closed circuit is given by $\mathbf{V_{emf} = \oint_L \bar{E} \cdot d\bar{l}}$

The magnetic flux ψ passing through a particular area is given by $\boldsymbol{\psi = \int_S \bar{B} \cdot d\bar{S}}$

From the above equations, we have

$$\oint_L \bar{E} \cdot d\bar{l} = -\frac{d}{dt}\int_S \bar{B} \cdot d\bar{S}$$

$$\boxed{\oint_L \bar{E} \cdot d\bar{l} = -\int_S \frac{\partial \bar{B}}{\partial t} \cdot d\bar{S}}$$

This is one of the Maxwell's equations for time-varying fields in integral form.

Using stokes theorem the above equation

$$\int_S \nabla \times \bar{E} \cdot d\bar{S} = -\int_S \frac{\partial \bar{B}}{\partial t} \cdot d\bar{S}$$

By eliminating the surface integrals,

$$\boxed{\nabla \times \bar{E} = -\frac{\partial \bar{B}}{\partial t}}$$

This is one of the Maxwell's equations for time-varying fields in differential form.

3. (a) Derive the expression for Energy density for steady magnetic field.

Ans:

The energy stored in the magnetic field is $\mathbf{W_m = \frac{1}{2} L I^2}$

Consider a differential volume in a magnetic field as shown in Fig.

Let the volume be covered with conducting sheets at the top and bottom surfaces with current ΔI.

The energy stored in the differential volume is $\mathbf{\Delta W_m = \frac{1}{2} \Delta L\, \Delta I^2}$

We know that $\mathbf{\Delta L = \frac{\Delta\psi}{\Delta I}}$

Therefore $\mathbf{\Delta W_m = \frac{1}{2} \Delta\psi \Delta I}$

Where

$$\mathbf{\Delta\psi = B\Delta x\Delta z}$$
$$\mathbf{\Delta I = H\Delta y}$$

Therefore

$$\mathbf{\Delta W_m = \frac{1}{2} BH\Delta x\Delta y\Delta z}$$

$$\mathbf{\Delta W_m = \frac{1}{2} BH\Delta V}$$

The energy density stored in the magnetic field is

$$\mathbf{w_m = \lim_{\Delta v \to 0} \frac{\Delta W_m}{\Delta V} = \frac{1}{2} BH}$$

$$\boxed{\mathbf{w_m = \frac{1}{2} BH = \frac{1}{2} \mu H^2}}$$

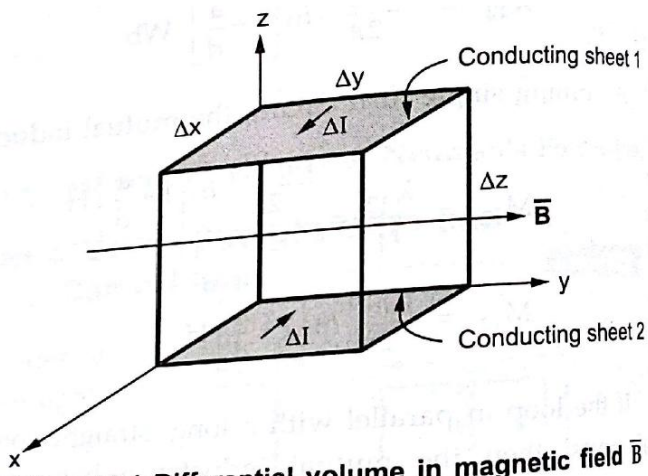


Fig. 8.13.1 Differential volume in magnetic field $\bar{B}$

3. (b) Distinguish the scalar and vector magnetic potential.

Ans:

Magnetic Scalar Potential:

The magnetic scalar potential V_m (in amperes) is defined as its negative gradient gives the magnetic field intensity.

Therefore $\boxed{\bar{\mathbf{H}} = -\nabla \mathbf{V_m}}$ **If** $\bar{\mathbf{J}} = \mathbf{0}$

We know that

$$\bar{\mathbf{J}} = \nabla \times \bar{\mathbf{H}} = \nabla \times (-\nabla \mathbf{V_m}) = \mathbf{0}$$

Because the curl of a gradient is zero.

Thus the magnetic scalar potential V_m is defined only in a region where J = 0.

The magnetic scalar potential between two points a and b is $\mathbf{V_{ma,b}} = \int_a^b \bar{\mathbf{H}} \cdot \mathbf{d}\bar{\mathbf{l}}$

Vector Magnetic Potentials:

The vector magnetic potential A (in Wb/m) is defined as its curl gives the magnetic flux density.

$$\bar{\mathbf{B}} = \nabla \times \bar{\mathbf{A}}$$

The vector magnetic potential can also be expressed as

$$\boxed{\bar{\mathbf{A}} = \int_L \frac{\mu_0 \mathbf{I} \mathbf{d}\bar{\mathbf{l}}}{4\pi \mathbf{R}}} \quad \text{for line current}$$

The magnetic flux passing through a given area is given by

$$\boldsymbol{\psi} = \oint_L \bar{\mathbf{A}} \cdot \mathbf{d}\bar{\mathbf{l}}$$

4. Describe the inconsistency in Ampere's law? How it is rectified by Maxwell?

Ans:

According to ampere's law, for static EM fields $\nabla \times \bar{\mathbf{H}} = \bar{\mathbf{J}} \quad ----(\mathbf{1})$

Taking divergence on the both sides of the above equation, we get

$$\nabla \cdot (\nabla \times \bar{\mathbf{H}}) = \nabla \cdot \bar{\mathbf{J}}$$

But the divergence of the curl of any vector field is identically zero, hence

$$\nabla \cdot (\nabla \times \bar{\mathbf{H}}) = \mathbf{0} = \nabla \cdot \bar{\mathbf{J}} \quad -----(\mathbf{2})$$

But the continuity equation for time varying fields is given by

$$\nabla \cdot \bar{\mathbf{J}} = -\frac{\partial \rho_v}{\partial t} \neq \mathbf{0} \quad -------(\mathbf{3})$$

Thus eqs.(2) and (3) are obviously incompatible for time-varying conditions. In other words Ampere's law is not consistent with time varying fields and needs modification to make it consistent for time varying fields. To do this, we add a term to $\bar{\mathbf{J}}_\mathbf{d}$ eq. (1) so that it becomes.

$$\nabla \times \bar{\mathbf{H}} = \bar{\mathbf{J}} + \bar{\mathbf{J}}_\mathbf{d} \quad -------(\mathbf{4})$$

Taking divergence on the both sides, we get

$$\nabla \cdot (\nabla \times \bar{\mathbf{H}}) = \nabla \cdot (\bar{\mathbf{J}} + \bar{\mathbf{J}}_\mathbf{d})$$

Again, the divergence of the curl of any vector is zero. Hence:

$$\nabla \cdot (\bar{\mathbf{J}} + \bar{\mathbf{J}}_\mathbf{d}) = \mathbf{0}$$

$$\nabla \cdot \bar{\mathbf{J}} + \nabla \cdot \bar{\mathbf{J}}_\mathbf{d} = \mathbf{0}$$

$$\nabla \cdot \bar{\mathbf{J}}_\mathbf{d} = -\nabla \cdot \bar{\mathbf{J}}$$

But from the continuity equation $\nabla \cdot \bar{J} = -\frac{\partial \rho_v}{\partial t}$

Therefore $\nabla \cdot \bar{J}_d = \frac{\partial \rho_v}{\partial t}$

But from Gauss's law $\nabla \cdot \bar{D} = \rho_v$

Therefore $\nabla \cdot \bar{J}_d = \frac{\partial}{\partial t}(\nabla \cdot \bar{D})$

$$\nabla \cdot \bar{J}_d = \nabla \cdot \frac{\partial \bar{D}}{\partial t}$$

Removing Divergence on both sides

$$\bar{J}_d = \frac{\partial \bar{D}}{\partial t} \quad ------(5)$$

Substitute equation (5) in equation (4) , we get

$$\nabla \times \bar{H} = \bar{J} + \frac{\partial \bar{D}}{\partial t}$$

This is Maxwell's equation (based on Ampere's circuit law) for a time-varying field.
The term $\bar{J}_d = \frac{\partial \bar{D}}{\partial t}$ is known as displacement current density and $\bar{J} = \sigma \bar{E}$ is the conduction current density.

5. Derive the four Maxwell's equations both in integral and differential forms.

Ans:

1. FARADAY'S LAW: Faraday law states that the induced emf V_{emf} (in volts), in any closed circuit is equal to the time rate of change of the magnetic flux linkage by the circuit.

$$V_{emf} = -\frac{d\psi}{dt}$$

Induced e.m.f in the closed circuit is given by $V_{emf} = \oint_L \bar{E} \cdot d\bar{l}$

The magnetic flux ψ passing through a particular area is given by $\psi = \int_S \bar{B} \cdot d\bar{S}$

From the above equations, we have

$$\oint_L \bar{E} \cdot d\bar{l} = -\frac{d}{dt}\int_S \bar{B} \cdot d\bar{S}$$

$$\boxed{\oint_L \bar{E} \cdot d\bar{l} = -\int_S \frac{\partial \bar{B}}{\partial t} \cdot d\bar{S}}$$

This is one of the Maxwell's equations for time-varying fields in integral form.
Using stokes theorem the above equation

$$\int_S \nabla \times \bar{E} \cdot d\bar{S} = -\int_S \frac{\partial \bar{B}}{\partial t} \cdot d\bar{S}$$

By eliminating the surface integrals,

$$\boxed{\nabla \times \bar{E} = -\frac{\partial \bar{B}}{\partial t}}$$

This is one of the Maxwell's equations for time-varying fields in differential form.

2. Inconsistency Of Ampere's Law (Modified Ampere's Law) :

According to ampere's law, for static EM fields $\nabla \times \bar{H} = \bar{J} \quad ----(1)$

Taking divergence on the both sides of the above equation, we get

$$\nabla \cdot (\nabla \times \bar{H}) = \nabla \cdot \bar{J}$$

But the divergence of the curl of any vector field is identically zero, hence

$$\nabla \cdot (\nabla \times \bar{H}) = 0 = \nabla \cdot \bar{J} \quad ------(2)$$

But the continuity equation for time varying fields is given by

$$\nabla \cdot \bar{J} = -\frac{\partial \rho_V}{\partial t} \neq 0 \qquad -------(3)$$

Thus eqs.(2) and (3) are obviously incompatible for time-varying conditions. In other words Ampere's law is not consistent with time varying fields and needs modification to make it consistent for time varying fields. To do this, we add a term to $\bar{J}_d$ eq. (1) so that it becomes.

$$\nabla \times \bar{H} = \bar{J} + \bar{J}_d \quad -------(4)$$

Taking divergence on the both sides, we get

$$\nabla \cdot (\nabla \times \bar{H}) = \nabla \cdot (\bar{J} + \bar{J}_d)$$

Again, the divergence of the curl of any vector is zero. Hence:

$$\nabla \cdot (\bar{J} + \bar{J}_d) = 0$$

$$\nabla \cdot \bar{J} + \nabla \cdot \bar{J}_d = 0$$

$$\nabla \cdot \bar{J}_d = -\nabla \cdot \bar{J}$$

But from the continuity equation $\nabla \cdot \bar{J} = -\frac{\partial \rho_V}{\partial t}$

Therefore $\nabla \cdot \bar{J}_d = \frac{\partial \rho_V}{\partial t}$

But from Gauss's law $\nabla \cdot \bar{D} = \rho_V$

Therefore $\nabla \cdot \bar{J}_d = \frac{\partial}{\partial t}(\nabla \cdot \bar{D})$

$$\nabla \cdot \bar{J}_d = \nabla \cdot \frac{\partial \bar{D}}{\partial t}$$

Removing Divergence on both sides

$$\bar{J}_d = \frac{\partial \bar{D}}{\partial t} \quad ------(5)$$

Substitute equation (5) in equation (4) , we get

$$\nabla \times \bar{H} = \bar{J} + \frac{\partial \bar{D}}{\partial t}$$

This is Maxwell's equation (based on Ampere's circuit law) for a time-varying field.

<u>3. Gauss's law:</u>

Gauss's law states that the total electric flux ψ passing through any closed surface is equal to the total charge enclosed by that surface.

$$\boxed{\psi = Q}$$

$$\text{Total flux } \psi = \oint_s \bar{D} \cdot d\bar{s}$$

$$\text{Total charge enclosed } Q = \int_v \rho_v dv$$

$$\boxed{\oint_s \bar{D} \cdot \overline{ds} = \int_v \rho_v dv}$$

By applying divergence theorem to the first term in the above equation

$$\oint_s \bar{D} \cdot \overline{ds} = \int_v \nabla \cdot \bar{D} dv$$

$$\int_v \nabla \cdot \bar{D} dv = \int_v \rho_v dv$$

$$\boxed{\nabla \cdot \bar{D} = \rho_v}$$

This is another Maxwell's equation.

4. Non existing of isolated magnetic poles:

The magnetic flux lines always form closed loops as shown in Fig.(b). This is due to the fact that it is not possible to have isolated magnetic poles.

Thus the total flux through a closed surface in a magnetic field must be zero as shown in fig. That is

$$\boxed{\oint \bar{B} \cdot d\bar{S} = 0} \qquad \text{----------} \ (1)$$

This equation refers to the law of conservation of magnetic flux.

By applying the divergence theorem to eq. (1), we obtain

$$\oint \bar{B} \cdot d\bar{S} = \int_V \nabla \cdot \bar{B} dV = 0$$

Or $\boxed{\nabla \cdot \bar{B} = 0}$

This equation is the another Maxwell's equation.

6. (a) Write down the Maxwell's equations in word statement.

1st equation:

$$\oint_s \bar{D} \cdot d\bar{S} = \int_V \rho_V dV$$

Statement: "The total electric flux ψ passing through any closed surface is equal to the total charge enclosed by that surface".

2nd equation:

$$\oint_s \bar{B} \cdot d\bar{S} = 0$$

Statement: "The surface integral of magnetic flux density over a closed surface is always equal to zero"

3rd equation:

$$\oint_L \bar{E} \cdot d\bar{l} = -\int_S \frac{\partial \bar{B}}{\partial t} \cdot d\bar{S}$$

Statement: "The line integral of electromotive force (emf) around any closed path is equal to the negative surface integral of time rate of change of magnetic flux density over the surface bounded by the path".

4th equation:

$$\oint_L \bar{H} \cdot d\bar{l} = \int_S \left(\bar{J} + \frac{\partial \bar{D}}{\partial t} \right) \cdot d\bar{S}$$

Statement: "The line integral of magneto motive force (mmf) around any closed path is equal to the surface integral of conduction and displacement current densities over the surface bounded by the path".

6. (b) A charge of 12 C has velocity of $5a_x + 2a_y - 3a_z$ m/s. Determine 'F' on the charge in the field of $B = 4a_x + 4a_y + 3a_z Wb/m^2$.

Ans:

Given that $Q = 12C$

$\bar{u} = 5\bar{a}_x + 2\bar{a}_y - 3\bar{a}_z$ m/s

$\bar{B} = 4\bar{a}_x + 4\bar{a}_y + 3\bar{a}_z$ wb/m²

The force $\bar{F}$ on the charge in the field is given by

$$\bar{F} = Q(\bar{u} \times \bar{B}) \rightarrow (1)$$

$$\bar{u} \times \bar{B} = \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ 5 & 2 & -3 \\ 4 & 4 & 3 \end{vmatrix}$$

$$= (6+12)\bar{a}_x - (15+12)\bar{a}_y + (20-8)\bar{a}_z$$

$$= 18\bar{a}_x - 27\bar{a}_y + 12\bar{a}_z$$

Substitute above in Eq (1),

$$\bar{F} = 12(18\bar{a}_x - 27\bar{a}_y + 12\bar{a}_z)$$

$$= 216\bar{a}_x - 324\bar{a}_y + 144\bar{a}_z \text{ N}$$

$$\boxed{\therefore \bar{F} = 216\bar{a}_x - 324\bar{a}_y + 144\bar{a}_z \text{ N}}$$

Short Answer Questions

UNIT-I

1. Define Electric field and Electric flux density.

Ans:

Electric field:

An **electric field** is a region of space around an electrically charged object in which other charged objects experience a force. It is a vector field that represents the force per unit positive charge at any point in space.

Electric Flux Density(D):

The total electric flux per unit area is defined as the electric flux density.

2. State and explain the Coulomb's law.

Ans:

- **Coulomb's law** states that the force F between two point charges Q_1 and Q_2 is:

  1. Along the line joining them

  2. Directly proportional to the product Q_1Q_2 of the charges

  3. Inversely proportional to the square of the distance *R* between them

Expressed mathematically,

$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_o R^2}$$

3. Define the convection and conduction currents.

Ans:

Convection current:

Convection current occurs when current flows through an insulating medium such as liquid, rarefied gas, or a vacuum. A beam of electrons in a vacuum tube, for example, is a convection current.

Conduction current:

Conduction current is defined as the current produced due to flow of electrons in conductors on the application of electric field.

4. List the limitations of Gauss's law.

Ans:

- It is useful only for symmetric charge distribution.
- It can applicable only for closed surfaces.
- It can applicable only if the surface under consideration encloses the entire charge.

5. What do you mean by conservative field?

Ans:

A field whose line integral does not depend on the path of integration is called conservative field. Electrostatic field is conservative field.

UNIT-II

1. State Ampere's circuital law.

Ans:

Ampere's circuit law states that the line integral of H around a closed path is the same as the net current I enclosed by the path.

$$\boxed{\oint \bar{\mathbf{H}}.\mathbf{d}\bar{\mathbf{l}} = \mathbf{I}} \qquad \text{---------- (1)}$$

2. Explain briefly about Magnetic Flux Density.

Ans:

The total magnetic flux passing through unit normal area is called magnetic flux density. It is denoted by **B**, whose units are Wb/m^2 or Tesla (T).

3. State Biot-Savarts Law.

Ans:

Biot-Savart's law states that the magnetic field intensity dH produced at a point P, as shown in Fig by the differential current element Idl is proportional to the product Idl and the sine of the angle α between the element and the line joining P to the element and is inversely proportional to the square of the distance R between P and the element.

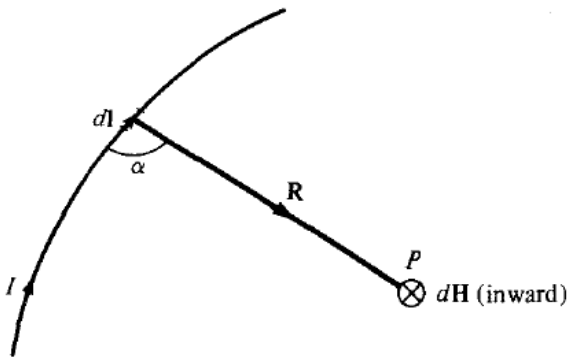


Fig. Magnetic field dH at P due to current element I dl.

That is $$\mathbf{dH} \propto \frac{\mathbf{Idl}\ \sin\alpha}{\mathbf{R^2}}$$

$$\mathbf{dH} = \mathbf{K}\frac{\mathbf{Idl}\ \sin\alpha}{\mathbf{R^2}}$$

Where k is the constant of proportionality.

In SI units, k = l/4π, so above equation becomes

$$\mathbf{dH} = \frac{\mathbf{Idl}\ \sin\alpha}{\mathbf{4\pi R^2}}$$

4. Write Maxwell's equations for time varying fields.

Ans:

$$1.\ \nabla \cdot \bar{\mathbf{D}} = \rho_V$$

$$2.\ \nabla \cdot \bar{\mathbf{B}} = 0$$

$$3.\ \nabla \times \bar{\mathbf{E}} = -\frac{\partial \bar{\mathbf{B}}}{\partial t}$$

$$4.\ \nabla \times \bar{\mathbf{H}} = \bar{\mathbf{J}} + \frac{\partial \bar{\mathbf{D}}}{\partial t}$$

5. List the two Maxwell's equations for magnetostatic filed.

Ans:

1. From Ampere's circuit law

$$\boxed{\oint \bar{\mathbf{H}}.\mathbf{d}\bar{\mathbf{l}} = \mathbf{I}}$$

$$\boxed{\nabla \times \bar{\mathbf{H}} = \bar{\mathbf{J}}}$$

2. From the law of conservation of magnetic flux.

$$\boxed{\oint \bar{\mathbf{B}} \cdot \mathbf{d}\bar{\mathbf{S}} = \mathbf{0}}$$

$$\boxed{\boldsymbol{\nabla} \cdot \bar{\mathbf{B}} = \mathbf{0}}$$